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ABSTRACT

Understanding the properties of warm dense hydrogen is of key importance for the modeling of compact astrophysical objects and to understand and further optimize inertial confinement fusion applications. The workhorse of warm dense matter theory is thermal density functional theory (DFT), which, however, suffers from two limitations: (i) its accuracy can depend on the utilized exchange–correlation functional, which has to be approximated, and (ii) it is generally limited to single-electron properties such as the density distribution. Here, we present a new ansatz combining time-dependent DFT results for the dynamic structure factor $S_{ee}(\mathbf{q}, \omega)$ with static DFT results for the density response. This allows us to estimate the electron–electron static structure factor $S_{ee}(\mathbf{q})$ of warm dense hydrogen with high accuracy over a broad range of densities and temperatures. In addition to its value for the study of warm dense matter, our work opens up new avenues for the future study of electronic correlations exclusively within the framework of DFT for a host of applications.

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I. INTRODUCTION

Over the last decade, there has been a surge of interest in the properties of matter at extreme densities (Wigner–Seitz radius $r_s = d/a_B \sim 1$), temperatures ($\Theta = k_B T/E_F \sim 1$, with E_F the electronic Fermi temperature¹) and pressures ($P \sim 1\text{--}10^4$ MBar).² Such warm dense matter (WDM) conditions abound in a variety of compact astrophysical objects,^{3–5} such as giant planet interiors.^{6,7} The high-pressure, moderate-temperature regime is of great interest for materials synthesis and discovery, with the demonstration of the laser-driven formation of nanodiamonds by Kraus *et al.*^{8,9} being a case in point. A particularly intriguing technological application

is given by inertial confinement fusion (ICF),^{10,11} where both the hydrogen fusion fuel and the ablator material have to traverse the WDM regime in a controlled way.¹² Indeed, there have been a number of very recent spectacular experimental achievements,^{13–15} but the systematic optimization of fusion yields that is required to make ICF an economical source of clean energy¹⁶ will require integrated modeling with real predictive capability.

From a theoretical perspective, WDM is characterized by the complex interplay of effects such as Coulomb coupling, quantum degeneracy, thermal excitation, and partial ionization,^{3,17,18} which have to be taken into account holistically; this is a formidable challenge even for state-of-the-art computational methods. Thermal

density functional theory (DFT)¹⁹ has emerged as the workhorse of modern WDM theory, but it has two main limitations: (i) its accuracy depends decisively on the electronic exchange–correlation (XC) functional, which has to be approximated and supplied as an external input; (ii) it is by construction, limited to single-electron observables such as the density distribution $n_e(\mathbf{r})$. There is a broad consensus in the community on the importance of (i), which has resulted in a number of new XC functionals that take into account temperature effects^{20–25} and in first attempts at benchmarking different functionals under WDM conditions.^{26–31} Here, we address point (ii) by introducing a new ansatz combining linear-response time-dependent DFT (TDDFT) results for the (inelastic) electronic dynamic structure factor $S_{ee}(\mathbf{q}, \omega)$ ³² with the static linear density response χ_{ee} computed from the direct perturbation method.^{28,31,33–35} This allows us to estimate the electron–electron static structure factor $S_{ee}(\mathbf{q})$, i.e., the Fourier transform of the pair correlation function $g_{ee}(\mathbf{r})$, without the need for any external input in addition to the usual XC functional. As a practical example, we consider warm dense hydrogen, which is of key importance for astrophysics and ICF applications alike,¹⁸ and find excellent agreement with highly accurate path integral Monte Carlo (PIMC) reference data^{36–38} where they are available. In addition, we present new results for lower temperatures, where PIMC simulations break down owing to the well-known fermion sign problem.^{39,40}

We expect our work to be of considerable interest for a gamut of future applications. First and foremost, we open up the way to study electronic correlations consistently within DFT on a true *ab initio* level. This is directly relevant to the interpretation of experiments with WDM where $S_{ee}(\mathbf{q})$ is probed both with X-ray diffraction and X-ray Thomson scattering diagnostics.^{34,41–44} In addition, our work might give new insights into the construction of advanced XC functionals based on combination of the adiabatic connection formula with the fluctuation–dissipation theorem.^{45,46} Finally, our setup has great potential to benchmark and constrain dynamic XC kernels^{47–50} and to guide novel developments in the field of TDDFT.

II. RESULTS

It has become a common practice to consider the imaginary-time correlation function (ITCF)^{34,51–56}

$$F_{ee}(\mathbf{q}, \tau) = \int_{-\infty}^{\infty} d\omega S_{ee}(\mathbf{q}, \omega) e^{-\tau\hbar\omega}, \quad (1)$$

which is given by the two-sided Laplace transform of the dynamic structure factor $S_{ee}(\mathbf{q}, \omega)$; the sought after static structure factor is given by $S_{ee}(\mathbf{q}) = F_{ee}(\mathbf{q}, 0)$. Next, we decompose the full dynamic structure factor into an elastic and inelastic part,

$$S_{ee}(\mathbf{q}, \omega) = \underbrace{W_R(\mathbf{q})\delta(\omega)}_{S_{el}(\mathbf{q}, \omega)} + S_{inel}(\mathbf{q}, \omega), \quad (2)$$

with $W_R(\mathbf{q})$ being the Rayleigh weight^{57,58} characterizing the electronic localization around the nuclei; note that Eq. (2) exclusively relies on the higher mass of the nuclei compared with that of the electrons and does not presuppose an additional decomposition into bound and free electrons.^{59,60} The Rayleigh weight constitutes a standard observable in DFT molecular dynamics calculations,^{57,58} and $S_{inel}(\mathbf{q}, \omega)$ is computed within the framework of linear-response

TDDFT using the efficient Liouville–Lanczos method³² with the PBE XC functional⁶¹ and the corresponding adiabatic XC kernel; see the [supplementary material](#) for additional details.

Evaluating Eq. (1) for $\tau = 0$ gives us the direct TDDFT results for $S_{ee}(\mathbf{q})$, which are shown as the red circles in [Fig. 1](#) for solid-density hydrogen ($\rho = 0.08 \text{ g/cm}^3$) at the electronic Fermi temperature of $T = 4.8 \text{ eV}$. These conditions can be realized in experiments with hydrogen jets^{62,63} and are also important for the initial phase of ICF experiments with cryogenic fuel.¹³ In addition, an ionization degree of $Z \sim 0.5$ has been reported in the literature,^{38,64,65} making these conditions particularly interesting. The black crosses in [Fig. 1](#) depict exact PIMC reference data,^{36,37} which provide an unassailable baseline for other theories. For completeness, we have also included corresponding results for a uniform electron gas (UEG)^{66,67} under the same conditions (solid black curve). For $q \lesssim 3q_F$, the UEG static structure factor exhibits a sharp drop due to the perfect screening in the one-component plasma;⁶⁸ by contrast, the PIMC results for hydrogen attain a finite value in the limit of $q \rightarrow 0$, in accordance with the compressibility sum rule.⁵⁷

Let us next consider the red circles and plusses in [Fig. 1](#), which show direct TDDFT results for $N = 32$ and 14 hydrogen atoms, respectively. We find no systematic finite-size effects, which is a general feature of q -resolved properties^{69–73} under these conditions. Overall, the direct TDDFT results capture the correct qualitative trend of $S_{ee}(\mathbf{q})$ for $q \lesssim 3q_F$, which is shaped by the electronic localization around the ions. At the same time, we find significant deviations over the entire range of wavenumbers (deviations with respect to PIMC are shown in the inset of [Fig. 1](#)), which are comparable in magnitude to a simpler chemical Chihara model;³⁸ see the dashed blue curve.

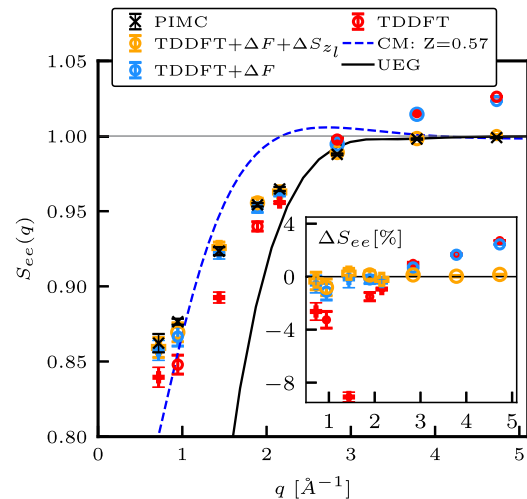


FIG. 1. Electronic static structure factor $S_{ee}(\mathbf{q})$ of hydrogen at $\rho = 0.08 \text{ g/cm}^3$ ($r_s = 3.23$) and $T = 4.8 \text{ eV}$ ($\Theta = 1$) for $N = 32$ (circles) and $N = 14$ (plusses) atoms. Black crosses, quasi-exact PIMC reference data;³⁷ red symbols, direct TDDFT; blue symbols, TDDFT + χ_A -correction [Eq. (4)]; yellow symbols, TDDFT + $\chi_A(\mathbf{q})$ and $\Delta\chi_i$ [Eq. (6)] corrections; dashed blue curve, chemical model³⁸ (see [supplementary material](#)). The inset shows the relative deviation of TDDFT datasets with respect to PIMC.

To understand the origin of the limited accuracy of the direct TDDFT results, we follow Ref. 32 and transform our TDDFT results for $S_{\text{inel}}(\mathbf{q}, \omega)$ to the τ domain via Eq. (1). A corresponding comparison against exact PIMC reference data^{36–38} for $F_{\text{inel}}(\mathbf{q}, \tau) = F_{ee}(\mathbf{q}, \tau) - W_R(\mathbf{q})$ and $F_{\text{el}}(\mathbf{q}, \tau) = W_R(\mathbf{q})$ is shown in Fig. 2(a) for $q = 1.89 \text{ \AA}^{-1}$. While PIMC and DFT results for the latter are basically indistinguishable, we find excellent agreement over the entire τ range for $F_{\text{inel}}(\mathbf{q}, \tau)$, except for a constant shift; this must be connected with the $\omega \rightarrow 0$ limit of $S_{\text{inel}}(\mathbf{q}, \omega)$,^{38,51} since any effect at finite ω would affect the τ dependence of $F(\mathbf{q}, \tau)$. The corresponding TDDFT results for $S_{\text{inel}}(\mathbf{q}, \omega)$ are shown in Fig. 2(b). The inset shows a magnified segment around the $\omega \rightarrow 0$ limit, which exhibits an unexpected drop for $\omega \lesssim 1 \text{ eV}$. This reflects the absence of a continuum (or quasi-continuum) portion of the spectra that contributes to thermal transitions with small energy differences, since TDDFT modeling has computational limitations in this regard (see the [supplementary material](#)).

To correct for this deficiency, we consider the imaginary-time version of the fluctuation–dissipation theorem,⁵¹ which connects the area under the ITCF with the linear density response to an external harmonic perturbation,

$$\chi_{\text{inel}}(\mathbf{q}) = -n_e \int_0^\beta d\tau F_{\text{inel}}(\mathbf{q}, \tau). \quad (3)$$

Clearly, accurate knowledge of Eq. (3) would be sufficient to correct for the constant shift observed in Fig. 2. Indeed, there is strong empirical evidence that equilibrium DFT simulations using the

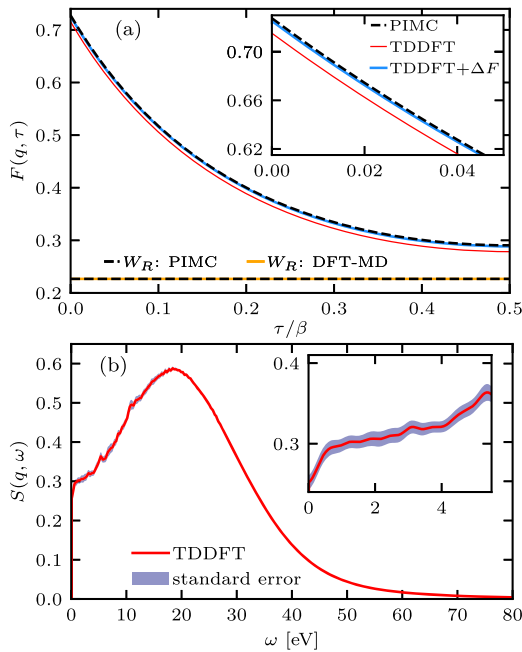


FIG. 2. (a) Inelastic ITCF $F_{\text{inel}}(\mathbf{q}, \tau)$ (top curves) and elastic ITCF $F_{\text{el}}(\mathbf{q}, \tau) = W_R(\mathbf{q})$ (horizontal lines) computed from PIMC (dashed black), direct TDDFT (solid red), and TDDFT + χ_A correction [Eq. (4)] (solid blue) for $T = 4.8 \text{ eV}$, $\rho = 0.08 \text{ g/cm}^3$, and $q = 1.892 \text{ \AA}^{-1}$. (b) Corresponding dynamic structure factor $S_{\text{inel}}(\mathbf{q}, \omega)$, with the inset showing a magnified segment around $\omega \rightarrow 0$.

direct perturbation approach,^{31,33} i.e., applying an external perturbation of wavenumber q and amplitude A [cf. Figs. 3(b) and 3(c)], is capable of reproducing PIMC reference data to a remarkable degree. In Fig. 3, we show $\chi_{\text{inel}}(\mathbf{q})$ and confirm the expected trends: the direct TDDFT results (red symbols) systematically deviate from the PIMC reference results (black crosses); by contrast, the direct perturbation results [$\chi_A(\mathbf{q})$, blue symbols] are in excellent agreement with PIMC for all wavenumbers. It is straightforward to estimate a corresponding correction term for both $S_{ee}(\mathbf{q})$ and the full ITCF:

$$\Delta F(\mathbf{q}) \equiv \Delta S_{ee}(\mathbf{q}) = -\frac{k_B T}{n_e} [\chi_A(\mathbf{q}) - \chi_{\text{inel}}^{\text{TDDFT}}(\mathbf{q})], \quad (4)$$

which follows from the static limit of the Fourier–Matsubara series expansion;⁷⁴ see the [supplementary material](#) for additional details. The full electron–electron static structure factor is then given by

$$S_{ee}(\mathbf{q}) = W_R(\mathbf{q}) + S_{\text{inel}}^{\text{TDDFT}}(\mathbf{q}) + \Delta S_{ee}(\mathbf{q}), \quad (5)$$

combining the Rayleigh weight from DFT molecular dynamics, the integral over the inelastic dynamic structure factor from TDDFT, and the correction from a set of harmonically perturbed static DFT calculations. Analogous expressions can be derived, for example, for the full ITCF $F_{ee}(\mathbf{q}, \tau)$ and for the full static linear density response function $\chi_{ee}(\mathbf{q})$, see the blue curve in Fig. 2(a). Evaluating Eq. (5) leads to the blue circles in Fig. 1. They are in very good agreement with the PIMC results for $q \lesssim 3q_F$, but start to disagree at around three times the Fermi wavenumber. This is a direct consequence of the inconsistent combination of an adiabatic (i.e., static) XC kernel with the dynamic Kohn–Sham response function within linear-response TDDFT.⁶⁷ The correction in Eq. (4), utilizing the static density response, does not correct the inaccuracies due to the use of an adiabatic (static) XC kernel in the TDDFT calculations. While these inaccuracies are relatively mild for the DSF at a given individual wavenumber, the spurious increase in $S_{ee}(\mathbf{q})$ for large q accumulates for integrated properties such as the interaction energy.

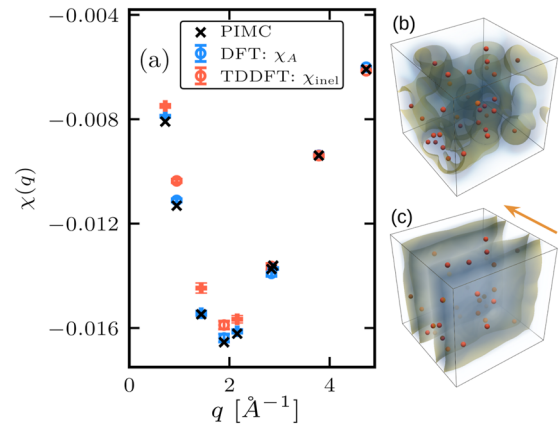


FIG. 3. (a) Static electron density response function $\chi_{\text{inel}}(\mathbf{q})$ [Eq. (3)] of hydrogen at $\rho = 0.08 \text{ g/cm}^3$ and $T = 4.8 \text{ eV}$. Black crosses, quasi-exact PIMC reference data;^{36,37} red symbols, direct TDDFT; blue symbols, direct perturbation approach, $\chi_A(\mathbf{q})$. (b) and (c) Electronic isosurfaces of the perturbed and unperturbed systems; see the [supplementary material](#) for details.

The underlying inconsistency has been investigated recently in detail for the UEG model by Dornheim *et al.*⁴⁹ in terms of the dynamic Matsubara density response function $\tilde{\chi}(\mathbf{q}, z_l)$, where $z_l = i2\pi k_B T l$ are the discrete bosonic imaginary Matsubara frequencies. To eliminate the effect of the static approximation in the XC kernel, we introduce a second correction using the Fourier–Matsubara expansion of the static structure factor. The familiar Fourier–Matsubara expansion for $S_{ee}(\mathbf{q})$ immediately leads to the first-order correction

$$\Delta\chi_l(\mathbf{q}) = -\frac{2k_B T}{n} \left\{ \sum_{l=1}^{\infty} [\tilde{\chi}_{\text{RPA}}(\mathbf{q}, z_l) - \tilde{\chi}_{\text{inel}}(\mathbf{q}, z_l)] \right\}, \quad (6)$$

which removes the effects of the XC kernel from the dynamic density response except for the static limit of $\tilde{\chi}_{\text{inel}}(\mathbf{q}, 0) = \chi_{\text{inel}}(\mathbf{q})$. Adding $\Delta\chi_l(\mathbf{q})$ to Eq. (5) leads to our final result for the electron static structure factor,

$$S_{ee}(\mathbf{q}) = W_R(\mathbf{q}) + S_{\text{inel}}^{\text{TDDFT}}(\mathbf{q}) + \Delta S_{ee}(\mathbf{q}) + \Delta\chi_l(\mathbf{q}). \quad (7)$$

The corresponding results are shown as the yellow symbols in Fig. 1 and are in striking agreement with the PIMC baseline over the entire range of wavenumbers. We stress that Eq. (7) does not require any external input apart from the usual electronic XC functional, as we explain in more detail in the [supplementary material](#).

To demonstrate the versatility of our approach, we next investigate $S_{ee}(\mathbf{q})$ at a substantially higher density of $\rho = 0.33 \text{ g/cm}^3$ over a broad range of temperatures in Fig. 4. The colored symbols show our fully corrected TDDFT results, i.e., Eq. (7). At the electronic Fermi temperature of $T = 12.5 \text{ eV}$, we again find excellent agreement with quasi-exact PIMC reference data³⁷ (black crosses) without the need for any empirical parameters. For lower temperatures, PIMC becomes unavailable because of the notorious fermion sign problem.^{39,40} Overall, we again find an increase of $S_{ee}(\mathbf{q})$ of hydrogen compared with the UEG model (solid curves) for $q \lesssim 3q_F$ due to electronic localization around the ions. Interestingly, our

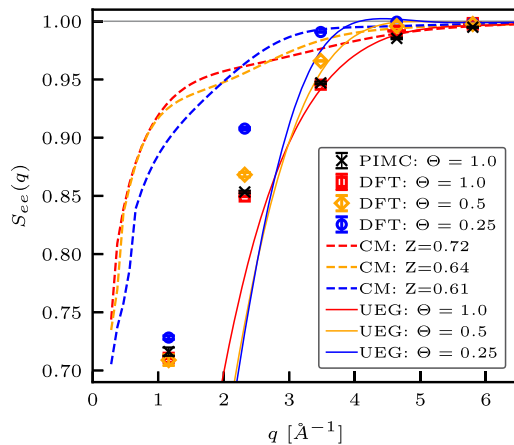


FIG. 4. Electron static structure factor $S_{ee}(\mathbf{q})$ of hydrogen at $\rho = 0.33 \text{ g/cm}^3$ ($r_s = 2$) and $T = 12.5 \text{ eV}$ ($\Theta = 1$, red), $T = 6.27 \text{ eV}$ ($\Theta = 0.5$, yellow), and $T = 3.13 \text{ eV}$ ($\Theta = 0.25$, blue). Symbols show fully corrected TDDFT data [Eq. (7)], and the curves have been computed on the basis of a chemical model;³⁸ see the [supplementary material](#) for details. The black crosses show quasi-exact PIMC reference data³⁷ for $\Theta = 1$.

TDDFT results for $S_{ee}(\mathbf{q})$ increase with decreasing temperature for intermediate wavenumber, owing to increasing electronic localization, whereas we find the opposite trend for the UEG. In addition, we compare our new results with those of a Chihara model (dashed lines), which, however, drastically overestimates the effect of localization for all temperatures. Taken together, Figs. 1 and 4 thus also clearly highlight the importance of *ab initio* simulations for the description of WDM in general, and for the interpretation of X-ray scattering experiments^{41,42,58} in particular.

III. DISCUSSION

In this work, we have introduced a new framework for the computation of the electron static structure factor $S_{ee}(\mathbf{q})$ from thermal DFT simulations. This has been achieved by combining the Rayleigh weight $W_R(\mathbf{q})$, the static linear density density response $\chi_{\text{inel}}(\mathbf{q})$ from a harmonically perturbed equilibrium calculation, and the frequency dependence from the dynamic Kohn–Sham response function. In addition, we remove the spurious effect of an adiabatic XC kernel on $S_{ee}(\mathbf{q})$ with a dynamic Matsubara correction, Eq. (6). The DFT results thus computed for $S_{ee}(\mathbf{q})$ are in excellent agreement with quasi-exact PIMC reference data, where they are available, without the need for any free empirical parameters. In addition, we have presented DFT results for hydrogen at significantly lower temperatures, where PIMC is unavailable because of the sign problem.

We are convinced that our work opens up a variety of avenues for important future research across various fields. First and foremost, we stress that the present investigation of warm dense hydrogen is important in its own right. For example, we have unambiguously demonstrated the substantial and systematic deficiency of ubiquitous chemical models, which constitute a widely used method for the interpretation of X-ray scattering experiments with WDM,^{75–78} for different densities and temperatures.

The results presented here were computed using the PBE XC functional. Generally, the methodology described can be applied in conjunction with any other XC functional. Depending on the type of material and the thermodynamic conditions, it may be necessary to use higher-rank XC functionals. Therefore, the limitations of the present method are defined by the ability of the chosen XC functional to describe both electronic and atomic structures accurately. For instance, describing dense hydrogen in regions where molecular formations are predominant is expected to be more challenging than the case considered here, which involves a significant degree of ionization.

Beyond hydrogen, we note that $S_{ee}(\mathbf{q})$ constitutes an observable in both X-ray diffraction and X-ray Thomson scattering experiments;^{41–43} our new DFT setup will thus be useful to predict the outcome of future experiments and to facilitate the interpretation of existing measurements.^{42,44,58} Indeed, a key advantage of DFT over the potentially more accurate but computationally more expensive PIMC approach is that it is available for a host of heavier elements and composite materials, and over a broad range of densities and temperatures.

Finally, we highlight the new opportunities for further methodological developments. For example, accurate DFT results for $S_{ee}(\mathbf{q})$ can be used to separately estimate the interaction energy W and, in

this way, to split the total energy into its potential and kinetic contributions. In addition, $S_{ee}(\mathbf{q})$ and W constitute important ingredients for the construction of advanced XC functionals based on the adiabatic connection formula.⁷⁹ Finally, it will be straightforward to combine the present approach with existing and upcoming dynamic XC kernels both in the real-frequency and imaginary Matsubara frequency domains.

SUPPLEMENTARY MATERIAL

Details of the simulation methods and parameters used are provided in the [supplementary material](#).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zhandos A. Moldabekov: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (equal); Visualization (lead); Writing – original draft (equal). **Xuecheng Shao:** Data curation (supporting); Formal analysis (supporting); Methodology (equal); Software (equal); Writing – original draft (supporting). **Hannah M. Bellenbaum:** Data curation (supporting); Formal analysis (supporting); Software (supporting); Writing – original draft (supporting). **Cheng Ma:** Data

curation (supporting); Formal analysis (supporting); Software (supporting); Writing – original draft (supporting). **Wenhui Mi:** Data curation (supporting); Formal analysis (supporting); Software (supporting); Writing – original draft (supporting). **Sebastian Schwalbe:** Data curation (supporting); Formal analysis (supporting); Methodology (supporting); Software (supporting); Writing – original draft (supporting). **Jan Vorberger:** Data curation (supporting); Formal analysis (supporting); Methodology (supporting); Software (supporting); Writing – original draft (supporting). **Tobias Dornheim:** Data curation (equal); Formal analysis (equal); Funding acquisition (lead); Investigation (equal); Methodology (equal); Resources (lead); Software (equal); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in RODARE at <https://doi.org/10.14278/rodare.4152>.

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